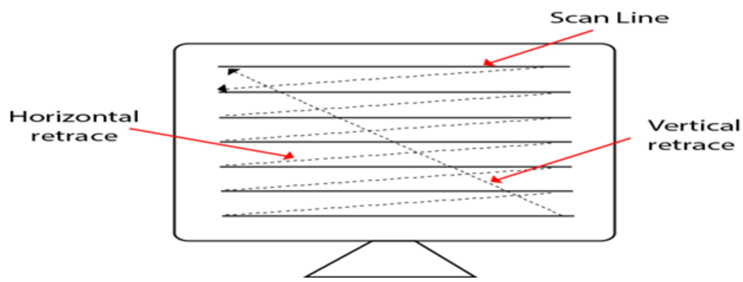
1. **Explain with diagram raster scan display technique. Differentiate between Vector scan display and Raster scan display.**

A Raster Scan Display is based on intensity control of pixels in the form of a rectangular box called Raster on the screen. Information of on and off pixels is stored in refresh buffer or Frame buffer. Televisions in our house are based on Raster Scan Method. The raster scan system can store information of each pixel position, so it is suitable for realistic display of objects. Raster Scan provides a refresh rate of 60 to 80 frames per second.

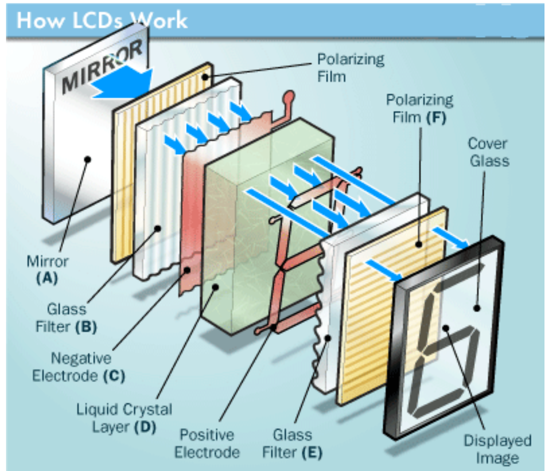


Vector Scan	Raster Scan
1. It has high Resolution	1. Its resolution is low.
2. It is more expensive	2. It is less expensive
3. Any modification if needed is easy	3.Modification is tough
4. Solid pattern is tough to fill	4.Solid pattern is easy to fill
5. Refresh rate depends on resolution	5. Refresh rate does not depend on the picture.
6. Only screen with view on an area is displayed.	6. Whole screen is scanned.

2. **Explain the principles of operation of each of the following. [Note: You may illustrate your answers with a diagram.]**

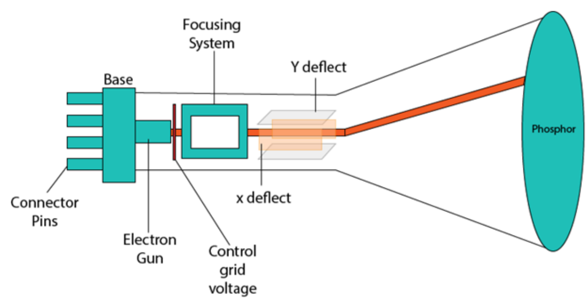
(i) Liquid crystal displays

The main principle behind liquid crystal molecules is that when an electric current is applied to them, they tend to untwist. This causes a change in the light angle passing through them. This causes a change in the angle of the top polarizing filter with respect to it. So little light is allowed to pass through that particular area of LCD. Thus that area becomes darker comparing to others.



(ii) Cathode Ray Tube displays

CRT stands for Cathode Ray Tube. CRT is a technology used in traditional computer monitors and televisions. The image on CRT display is created by firing electrons from the back of the tube of phosphorus located towards the front of the screen.
Once the electron heats the phosphorus, they light up, and they are projected on a screen. The color you view on the screen is produced by a blend of red, blue and green light.



Cathode Ray Tube

3. Consider the line from (0, 0) to (4, 6). Use DDA algorithm to rasterize this line.

Start point: (0,0), End point: (4,6).
dx = 4-0=4, dy = 6-0=6.
steps = max(|dx|, |dy|) = 6.
x increment = dx/steps = 4/6 ≈ 0.67.
y increment = dy/steps = 6/6 = 1.

Now we generate points:

Step	X (before rounding)	y (before rounding)	Plotted Pixel (rounded)
0	0.00	0.00	(0,0)
1	0.67	1.00	(1,1)
2	1.34	2.00	(1,2)
3	2.01	3.00	(2,3)

4	2.68	4.00	(3,4)
5	3.35	5.00	(3,5)
6	4.02	6.00	(4,6)

Now, draw the line from the above plotted pixel.

4. Define display processor. Which purposes does the storage tube serves? Describe the triad arrangement of red, green, and blue guns in color CRT.

A **display processor** is a special-purpose computer (or a graphics controller) whose main job is to relieve the main CPU from graphics tasks. It acts as an interface between the CPU and the display device (monitor/CRT).

The storage tube is a type of CRT display device (used in early graphics systems). Its purposes are:

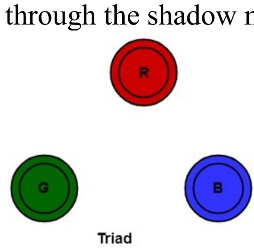
- **Permanent Display without Refreshing:** Once an image is drawn, it stays on the screen without the need for refreshing (unlike a normal CRT which needs continuous refresh).
- **Lower CPU and Memory Usage**: Since refreshing is not needed, no frame buffer or display processor is required.
- **Used for Static Images:** Suitable for applications like engineering drawings, weather maps, or text display where the image does not change frequently.
- **High Resolution:** Can display fine, detailed drawings because the image stays sharp until erased.

Triad arrangement of red, green, and blue guns in color CRT

The deflection system of the CRT operates on all 3 electron beams simultaneously; the 3 electron beams are deflected and focused as a group onto the shadow mask, which contains a sequence of holes aligned with the phosphor- dot patterns.

When the three beams pass through a hole in the shadow mask, they activate a dotted triangle, which occurs as a small color spot on the screen.

The phosphor dots in the triangles are organized so that each electron beam can activate only its corresponding color dot when it passes through the shadow mask.



5. What is scan conversion? Write down the advantage of developing algorithms for scan conversion.

It is a process of representing graphics objects a collection of pixels. The graphics objects are continuous. The pixels used are discrete. Each pixel can have either on or off state.

<u>Advantage of developing algorithms for scan conversion</u>

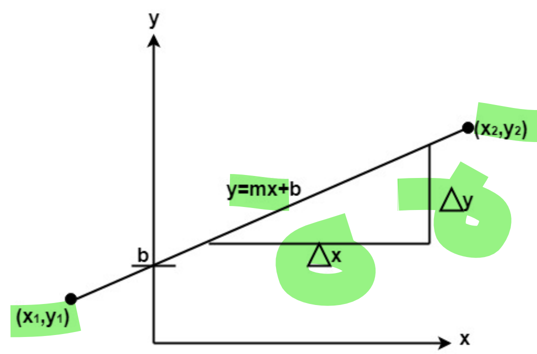
1. Algorithms can generate graphics objects at a faster rate.
2. Using algorithms memory can be used efficiently.

3. Algorithms can develop a higher level of graphical objects.

6. **Describe the process of scan conversion of a straight line. Write down the properties of a good line drawing algorithm.**

A straight line may be defined by two endpoints & an equation. In fig the two endpoints are described by (x_1,y_1) and (x_2,y_2). The equation of the line is used to determine the x, y coordinates of all the points that lie between these two endpoints.

Using the equation of a straight line, $y = mx + b$ where $m = dy/dx$ & b = the y interrupt, we can find values of y by incrementing x from $x = x_1$, to $x = x_2$. By scan-converting these calculated x, y values, we represent the line as a sequence of pixels.



Properties of Good Line Drawing Algorithm:

1. **Line should appear Straight:** We must appropriate the line by choosing addressable points close to it. If we choose well, the line will appear straight, if not, we shall produce crossed lines.
2. **Lines should terminate accurately:** Unless lines are plotted accurately, they may terminate at the wrong place.
3. **Lines should have constant density:** Line density is proportional to the no. of dots displayed divided by the length of the line.
4. **Line density should be independent of line length and angle:** This can be done by computing an approximating line-length estimate and to use a line-generation algorithm that keeps line density constant to within the accuracy of this estimate.

5. **Line should be drawn rapidly:** This computation should be performed by special-purpose hardware.

7. **If a line is drawn from (2, 3) to (6, 15) with use of DDA, how many points will be needed to generate such line?**

Step 1: Calculate differences
dx = x2 - x1 = 6 - 2 = 4
dy = y2 - y1 = 15 - 3 = 12

Step 2: Find number of steps
steps = max(|dx|, |dy|) = max(4, 12) = 12

Step 3: Find increments per step
x increment = dx / steps = 4 / 12 = 0.333
y increment = dy / steps = 12 / 12 = 1

Starting point: (x0, y0) = (2, 3)

Step	X (calculated)	Y (calculated)	Pixel Plotted (rounded)
0	2.000	3.000	(2, 3)
1	2.333	4.000	(2, 4)
2	2.667	5.000	(3, 5)
3	3.000	6.000	(3, 6)
4	3.333	7.000	(3, 7)
5	3.667	8.000	(4, 8)
6	4.000	9.000	(4, 9)
7	4.333	10.000	(4, 10)
8	4.667	11.000	(5, 11)
9	5.000	12.000	(5, 12)
10	5.333	13.000	(5, 13)
11	5.667	14.000	(6, 14)
12	6.000	15.000	(6, 15)

8. Derive the process of drawing a line using Bresenham's Line drawing algorithm with exact mathematical background.

1. Start at the first point $P_1(x_1, y_1)$.
2. Move horizontally towards the second point $P_2(x_2, y_2)$, one pixel at a time.
3. At each step, there are **two candidate pixels**:
   - **S (lower pixel)**: same y as current pixel.

$$x_{i+1} = x_i + 1, \quad y_{i+1} = y_i$$

   - **T (upper pixel)**: y increased by 1.

$$x_{i+1} = x_i + 1, \quad y_{i+1} = y_i + 1$$

4. Decide which pixel is closer using the **decision variable**:

$$d_i = 2\Delta y x_i - 2\Delta x y_i + c$$

where c is a constant based on the line equation.

Decision rule

- If $d_i < 0 \rightarrow$ choose **S** (lower pixel).
- If $d_i \geq 0 \rightarrow$ choose **T** (upper pixel).

Update rule for next step

- **If S is chosen**:

$$d_{i+1} = d_i + 2\Delta y$$

- **If T is chosen**:

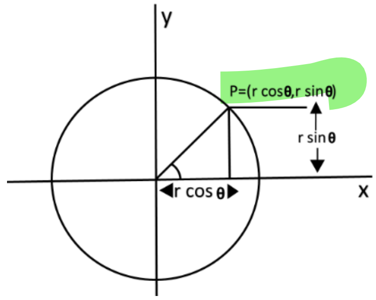
$$d_{i+1} = d_i + 2\Delta y - 2\Delta x$$

Initial decision variable

$$d_0 = 2\Delta y - \Delta x$$

9. Define a circle using polar co-ordinates. Write down the algorithm for drawing a circle using polar co-ordinates.

Defining a circle makes use of polar co-ordinates as shown in fig:



$x = r\cos\theta \qquad y = r\sin\theta$

Where θ=current angle

r = circle radius

x = x coordinate

y = y coordinate

By this method, θ is stepped from 0 to $\frac{\pi}{4}$ & each value of x & y is calculated.

<u>Algorithm:</u>

Step1: Set the initial variables:

r = circle radius

(h, k) = coordinates of the circle center

i = step size

θ_end=(22/7)/4

θ=0

Step2: If $\theta > \theta_{end}$ then stop.

Step3: Compute

$x = r * \cos\theta \qquad y = r * \sin\theta$

Step4: Plot the eight points, found by symmetry i.e., the center (h, k), at the current (x, y) coordinates.

Plot (x + h, y +k)	Plot (-x + h, -y + k)
Plot (y + h, x + k)	Plot (-y + h, -x + k)
Plot (-y + h, x + k)	Plot (y + h, -x + k)
Plot (-x + h, y + k)	Plot (x + h, -y + k)

Step5: Increment θ=θ+i

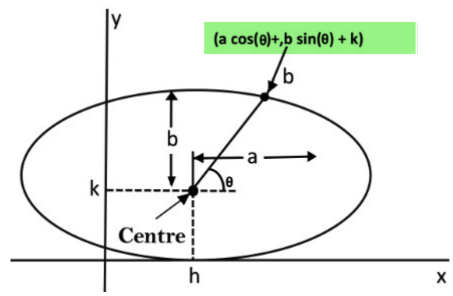
Step6: Go to step (ii).

10. Write down the drawback of polynomial method for scan conversion of an ellipse. Describe the trigonometric method for scan conversion of an ellipse.

Drawback:

1. This is an inefficient method.
2. It is not an interactive method for generating ellipse.
3. The table is required to see the trigonometric value.
4. Memory is required to store the value of θ.

The following equation defines an ellipse trigonometrically as shown in fig:



$x = a * \cos(\theta) + h$ and
$y = b * \sin(\theta) + k$
where (x, y) = the current coordinates
a = length of major axis
b = length of minor axis
θ= current angle
(h, k) = ellipse center

In this method, the value of θ is varied from 0 to $\frac{\pi}{2}$ radians. The remaining points are found by symmetry.

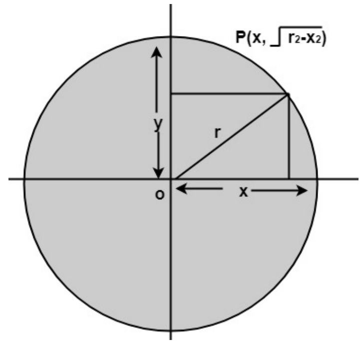
11. Write down the midpoint ellipse algorithm.

```
int x=0, y=b; [starting point]
int fx=0, fy=2a^2 b [initial partial derivatives]
int p = b^2-a^2 b+a^2/4
while (fx<="" {="" set="" pixel="" (x,="" y)="" x++;="" fx="fx" +="" 2b^2;
if (p<0)
p = p + fx +b2;
else
{
y--;
fy=fy-2a^2
p = p + fx +b^2-fy;
}
}
Setpixel (x, y);
p=b^2(x+0.5)^2+ a^2 (y-1)^2- a^2
b^2
while (y>0)
{
y--;
fy=fy-2a^2;
if (p>=0)
p=p-fy+a^2
else
{
x++;
fx=fx+2b^2
p=p+fx-fy+a^2;
}
Setpixel (x,y);
}
```

❖ **Defining a circle using Polynomial Method.**

The first method defines a circle with the second-order polynomial equation as shown in fig:

$y^2=r^2-x^2$



Where x = the x coordinate

y = the y coordinate

r = the circle radius

With the method, each x coordinate in the sector, from 90° to 45°, is found by stepping x from 0 to $\frac{r}{\sqrt{2}}$ & each y coordinate is found by evaluating $\sqrt{r^2-x^2}$ for each step of x.

Algorithm:

Step1: Set the initial variables

r = circle radius

(h, k) = coordinates of circle center

x=o

I = step size

$x_{end}=\frac{r}{\sqrt{2}}$

Step2: Test to determine whether the entire circle has been scan-converted.

If x > x_{end} then stop.

Step3: Compute y =$\sqrt{r^2-x^2}$

Step4: Plot the eight points found by symmetry concerning the center (h, k) at the current (x, y) coordinates.

Plot (x + h, y +k) Plot (-x + h, -y + k)
Plot (y + h, x + k) Plot (-y + h, -x + k)
Plot (-y + h, x + k) Plot (y + h, -x + k)
Plot (-x + h, y + k) Plot (x + h, -y + k)

Step5: Increment x = x + i

Step6: Go to step (ii).